Here are the detailed, step-by-step solutions for **Problems 35–48** from `1-riemann-sum.jpeg` using **Definition 5.4.3**:

$$A = \lim_{n \to \infty} \sum_{k=1}^{n} f(x_k^*)\Delta x \qquad where,\ \Delta x = \frac{b-a}{n}$$

Problems 35–40: Right Endpoints ($x_k^* = a + k\Delta x$)

35. $f(x) = \frac{x}{2}$ **over** $[1, 4]$

- **Parameters**: $a = 1, b = 4, \Delta x = \frac{4-1}{n} = \frac{3}{n}$.
- **Right Endpoint**: $x_k^* = 1 + \frac{3k}{n}$.
- **Function Value**: $f(x_k^*) = \frac{1}{2}\left(1 + \frac{3k}{n}\right) = \frac{1}{2} + \frac{3k}{2n}$.
- **Riemann Sum**:
  - $\sum_{k=1}^{n} f(x_k^*)\Delta x = \frac{3}{n}\left[\frac{1}{2}\sum_{k=1}^{n} 1 + \frac{3}{2n}\sum_{k=1}^{n} k\right] = \frac{3}{n}\left[\frac{n}{2} + \frac{3(n+1)}{4}\right] = \frac{15n+9}{4n}$
-
- **Area**:
  - $A = \lim_{n \to \infty} \frac{15n+9}{4n} = \frac{15}{4}$

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36. $f(x) = 5 - x$ **over** $[0, 5]$

- **Parameters**: $a = 0, b = 5, \Delta x = \frac{5}{n}$.
- **Right Endpoint**: $x_k^* = \frac{5k}{n}$.
- **Function Value**: $f(x_k^*) = 5 - \frac{5k}{n}$.
- **Riemann Sum**:
  - $\sum_{k=1}^{n} f(x_k^*)\Delta x = \frac{5}{n}\left[5n - \frac{5}{n}\frac{n(n+1)}{2}\right] = \frac{5}{n}\left[\frac{5n-5}{2}\right] = \frac{25(n-1)}{2n}$
-
- **Area**:
  - $A = \lim_{n \to \infty} \frac{25n-25}{2n} = \frac{25}{2}$

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37. $f(x) = 9 - x^2$ **over** $[0, 3]$

- **Parameters**: $a = 0, b = 3, \Delta x = \frac{3}{n}$.
- **Right Endpoint**: $x_k^* = \frac{3k}{n}$.
- **Function Value**: $f(x_k^*) = 9 - \frac{9k^2}{n^2}$.
- **Riemann Sum**:
- $\sum_{k=1}^{n} f(x_k^*)\Delta x = \frac{3}{n}\left[9n - \frac{9}{n^2}\frac{n(n+1)(2n+1)}{6}\right] = 27 - 9 - \frac{27}{2n} - \frac{9}{2n^2} = 18 - \frac{27}{2n} - \frac{9}{2n^2}$
-
- **Area**:
    - $A = \lim_{n \to \infty}\left(18 - \frac{27}{2n} - \frac{9}{2n^2}\right) = 18$

---

38. $f(x) = 4 - \frac{1}{4}x^2$ **over** $[0, 3]$

- **Parameters**: $a = 0, b = 3, \Delta x = \frac{3}{n}$.
- **Right Endpoint**: $x_k^* = \frac{3k}{n}$.
- **Function Value**: $f(x_k^*) = 4 - \frac{9k^2}{4n^2}$.
- **Riemann Sum**:
- $\sum_{k=1}^{n} f(x_k^*)\Delta x = \frac{3}{n}\left[4n - \frac{9}{4n^2}\frac{n(n+1)(2n+1)}{6}\right] = 12 - \frac{9}{4} - \frac{27}{8n} - \frac{9}{8n^2} = \frac{39}{4} - \frac{27}{8n} - \frac{9}{8n^2}$
-
- **Area**:
    - $A = \lim_{n \to \infty}\left(\frac{39}{4} - \frac{27}{8n} - \frac{9}{8n^2}\right) = \frac{39}{4}$

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39. $f(x) = x^3$ **over** $[2, 6]$

- **Parameters**: $a = 2, b = 6, \Delta x = \frac{4}{n}$.
- **Right Endpoint**: $x_k^* = 2 + \frac{4k}{n}$.
- **Function Value**: $f(x_k^*) = 8 + \frac{48k}{n} + \frac{96k^2}{n^2} + \frac{64k^3}{n^3}$.

- **Riemann Sum**:
  - $\sum_{k=1}^{n} f(x_k^*)\Delta x = \frac{4}{n}\left[8n + \frac{48}{n}\frac{n(n+1)}{2} + \frac{96}{n^2}\frac{n(n+1)(2n+1)}{6} + \frac{64}{n^3}\frac{n^2(n+1)^2}{4}\right]$
  - $= 320 + \frac{416}{n} + \frac{128}{n^2}$
- **Area**:
  - $A = \lim_{n\to\infty}\left(320 + \frac{416}{n} + \frac{128}{n^2}\right) = 320$

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40. $f(x) = 1 - x^3$ **over** $[-3, -1]$

- **Parameters**: $a = -3$, $b = -1$, $\Delta x = \frac{2}{n}$.
- **Right Endpoint**: $x_k^* = -3 + \frac{2k}{n}$.
- **Function Value**: $f(x_k^*) = 28 - \frac{54k}{n} + \frac{36k^2}{n^2} - \frac{8k^3}{n^3}$.
- **Riemann Sum**:
  - $\sum_{k=1}^{n} f(x_k^*)\Delta x = \frac{2}{n}\left[28n - \frac{54}{n}\frac{n(n+1)}{2} + \frac{36}{n^2}\frac{n(n+1)(2n+1)}{6} - \frac{8}{n^3}\frac{n^2(n+1)^2}{4}\right]$
  - $= 22 - \frac{26}{n} + \frac{8}{n^2}$
- **Area**:
  - $A = \lim_{n\to\infty}\left(22 - \frac{26}{n} + \frac{8}{n^2}\right) = 22$

Problems 41–44: Left Endpoints ($x_k^* = a + (k-1)\Delta x$)

41. $f(x) = \frac{x}{2}$ **over** $[1, 4]$

- **Parameters**: $a = 1$, $b = 4$, $\Delta x = \frac{3}{n}$.
- **Left Endpoint**: $x_k^* = 1 + \frac{3(k-1)}{n}$.
- **Riemann Sum**:

- $\sum_{k=1}^{n} f(x_k^*)\Delta x = \frac{3}{n}\left[\frac{n}{2} + \frac{3}{2n}\frac{(n-1)n}{2}\right] = \frac{15n-9}{4n}$
-
- **Area**:
  - $A = \lim_{n\to\infty} \frac{15n-9}{4n} = \frac{15}{4}$

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42. $f(x) = 5 - x$ **over** $[0, 5]$

- **Parameters**: $a = 0$, $b = 5$, $\Delta x = \frac{5}{n}$.
- **Left Endpoint**: $x_k^* = \frac{5(k-1)}{n}$.
- **Riemann Sum**:
  - $\sum_{k=1}^{n} f(x_k^*)\Delta x = \frac{5}{n}\left[5n - \frac{5}{n}\frac{(n-1)n}{2}\right] = \frac{25(n+1)}{2n}$
-
- **Area**:
  - $A = \lim_{n\to\infty} \frac{25n+25}{2n} = \frac{25}{2}$

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43. $f(x) = 9 - x^2$ **over** $[0, 3]$

- **Parameters**: $a = 0$, $b = 3$, $\Delta x = \frac{3}{n}$.
- **Left Endpoint**: $x_k^* = \frac{3(k-1)}{n}$.
- **Riemann Sum**:
  - $\sum_{k=1}^{n} f(x_k^*)\Delta x = \frac{3}{n}\left[9n - \frac{9}{n^2}\frac{(n-1)n(2n-1)}{6}\right] = 18 + \frac{27}{2n} - \frac{9}{2n^2}$
-
- **Area**:
  - $A = \lim_{n\to\infty} \left(18 + \frac{27}{2n} - \frac{9}{2n^2}\right) = 18$

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44. $f(x) = 4 - \frac{1}{4}x^2$ **over** $[0, 3]$

- **Parameters**: $a = 0$, $b = 3$, $\Delta x = \frac{3}{n}$.
- **Left Endpoint**: $x_k^* = \frac{3(k-1)}{n}$.
- **Riemann Sum**:
  - $\sum_{k=1}^{n} f(x_k^*)\Delta x = \frac{3}{n}\left[4n - \frac{9}{4n^2}\frac{(n-1)n(2n-1)}{6}\right] = \frac{39}{4} + \frac{27}{8n} - \frac{9}{8n^2}$
-
- **Area**:
  - $A = \lim_{n\to\infty}\left(\frac{39}{4} + \frac{27}{8n} - \frac{9}{8n^2}\right) = \frac{39}{4}$

Problems 45–48: Midpoints ($x_k^* = a + \left(k - \frac{1}{2}\right)\Delta x$)

45. $f(x) = 2x$ **over** $[0, 4]$

- **Parameters**: $a = 0$, $b = 4$, $\Delta x = \frac{4}{n}$.
- **Midpoint**: $x_k^* = \frac{4}{n}\left(k - \frac{1}{2}\right)$.
- **Riemann Sum**:
  - $\sum_{k=1}^{n} f(x_k^*)\Delta x = \frac{32}{n^2}\sum_{k=1}^{n}\left(k - \frac{1}{2}\right) = \frac{32}{n^2}\left(\frac{n^2}{2}\right) = 16$
-
- **Area**:
  - $A = \lim_{n\to\infty} 16 = 16$

---

46. $f(x) = 6 - x$ **over** $[1, 5]$

- **Parameters**: $a = 1$, $b = 5$, $\Delta x = \frac{4}{n}$.
- **Midpoint**: $x_k^* = 1 + \frac{4}{n}\left(k - \frac{1}{2}\right)$.
- **Riemann Sum**:
  - $\sum_{k=1}^{n} f(x_k^*)\Delta x = \frac{4}{n}\left[5n - \frac{4}{n}\left(\frac{n^2}{2}\right)\right] = \frac{4}{n}(3n) = 12$

- **Area**:
  - $A = \lim_{n \to \infty} 12 = 12$

---

47. $f(x) = x^2$ **over** $[0, 1]$

- **Parameters**: $a = 0$, $b = 1$, $\Delta x = \frac{1}{n}$.
- **Midpoint**: $x_k^* = \frac{1}{n}\left(k - \frac{1}{2}\right)$.
- **Riemann Sum**:
  - $\sum_{k=1}^{n} f(x_k^*)\Delta x = \frac{1}{n^3}\sum_{k=1}^{n}\left(k - \frac{1}{2}\right)^2 = \frac{1}{n^3}\left(\frac{n^3}{3} - \frac{n}{12}\right) = \frac{1}{3} - \frac{1}{12n^2}$
-
- **Area**:
  - $A = \lim_{n \to \infty}\left(\frac{1}{3} - \frac{1}{12n^2}\right) = \frac{1}{3}$

---

48. $f(x) = x^2$ **over** $[-1, 1]$

- **Parameters**: $a = -1$, $b = 1$, $\Delta x = \frac{2}{n}$.
- **Midpoint**: $x_k^* = -1 + \frac{2}{n}\left(k - \frac{1}{2}\right)$.
- **Riemann Sum**:
  - $\sum_{k=1}^{n} f(x_k^*)\Delta x = \frac{2}{n}\left[\frac{4}{n^2}\left(\frac{n^3}{3} - \frac{n}{12}\right) - 2n + n\right] = \frac{2}{3} - \frac{2}{3n^2}$
-
- **Area**:
  - $A = \lim_{n \to \infty}\left(\frac{2}{3} - \frac{2}{3n^2}\right) = \frac{2}{3}$